

A Multicenter Randomized Clinical Trial of Dapagliflozin in Patients Receiving Chronic Dialysis (The DARE-ESKD-2 Trial)



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Introduction: Dapagliflozin reduces heart failure (HF) incidence and hospitalizations in chronic kidney disease (CKD), but its efficacy and safety in patients on maintenance dialysis remain uncertain.

Methods: In this prospective, randomized, open-label, blinded end point trial performed at 3 dialysis clinics between January 2023 and September 2024, adults on chronic dialysis were randomized to once daily dapagliflozin 10 mg or standard care for 24 weeks. The primary end point was change in N-terminal pro-brain natriuretic peptide (NT-proBNP) levels. Secondary end points included changes in Kansas City Cardiomyopathy Questionnaire (KCCQ) scores and 6-minute walk test distance (6MWT). Exploratory analysis examined intergroup differences in echocardiographic measurements and laboratory tests. Safety analysis comprised adverse event rates.

Results: Of 110 screened individuals, 80 were consecutively assigned (mean age: 56 years; 54% male). The median (95% confidence interval [CI]) adjusted change in NT-proBNP was 13 (−64 to 281) pg/ml in the control group and −109 (−356 to 55) pg/ml in the dapagliflozin group. The intergroup difference in NT-proBNP change was not statistically significant after adjustment by baseline values (median [95% CI] difference: −155 [−327 to −33] pg/ml; $P = 0.065$). No difference was found between study arms regarding the least mean (95% CI) square difference (LSD) in KCCQ clinical summary score (7 points; $P = 0.073$), total summary score (7 points; $P = 0.094$), or 6MWT (−0.9 m; $P = 0.956$). Dapagliflozin did not increase the proportion of patients with improvement in 6MWT (34.5% vs. 40.0%; odds ratio [OR]: 1.27 [95% CI: 0.44–3.74]; $P = 0.657$) or KCCQ clinical summary score (18.8% vs. 22.9%; OR: 1.63 [95% CI: 0.33–9.12]; $P = 0.548$). No difference was found between study arms regarding echocardiographic measurements or safety end points.

Conclusion: Dapagliflozin was safe but did not affect surrogate markers of HF compared with standard care in patients receiving maintenance dialysis, suggesting no demonstrable cardiovascular benefit over 24 weeks in this population.

Kidney Int Rep (2026) 11, 106355; <https://doi.org/10.1016/j.ekir.2026.106355>

KEYWORDS: dapagliflozin; dialysis; heart failure

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Received 25 November 2025; revised 28 January 2026; accepted 3 February 2026; published online 10 February 2026

Sodium-glucose cotransporter 2 inhibitors (SGLT2i) have demonstrated consistent cardiovascular benefits, particularly in reducing hospitalizations for HF, among patients with or at risk for HF.^{1,2} These benefits have been observed in both HF with reduced ejection fraction and HF with preserved ejection fraction, as

shown in multiple large-scale cardiovascular outcome trials.² Meta-analyses and clinical trials have further shown that the cardiovascular benefits of SGLT2i are largely independent of baseline kidney function.³ In fact, in patients with advanced CKD, continuation of SGLT2i therapy is associated with lower risks of cardiovascular and kidney events than with discontinuation.^{4,5} Accordingly, SGLT2i is now considered a cornerstone of cardiovascular prevention in CKD.⁶

Patients receiving dialysis experience > 20-fold higher incidence of HF and cardiovascular death than the general population, yet no current therapies have demonstrated meaningful reduction of this risk.⁷ Hypothetically, cardiovascular benefit of SGLT2i therapy may occur in these patients via a large set of kidney-independent mechanisms. These include activation of transcriptional pathways that mimic nutrient deprivation and hypoxia and promoting ketosis and autophagy, thereby enhancing myocardial energetics.⁸ In addition, SGLT2i improve mitochondrial function by enhancing dynamics and bioenergetics while reducing reactive oxygen species production.⁹ They also improve endothelial function by upregulating endothelial nitric oxide synthase activity and attenuating oxidative stress, collectively facilitating vasodilation and reducing peripheral vascular resistance.^{10,11}

Despite this broad array of biologically plausible, kidney-independent mechanisms, whether the cardiovascular benefits of SGLT2i extend to patients on chronic dialysis remains a critical and unresolved clinical question. This study was designed to evaluate the safety of dapagliflozin and to test the hypothesis that it improves cardiac function, exercise capacity, and HF symptom burden in this high-risk population.

METHODS

The DARE-ESKD-2 trial was a prospective, randomized, open-label, blinded end point study performed between January 2023 and September 2024. The trial strictly followed the Good Clinical Practice guidelines, applicable regulatory requirements, and the Declaration of Helsinki principles. The institutional review board granted approval for each site before study initiation and all participants signed an informed consent form. The study protocol provided in the Supplementary Methods was previously published¹² and was registered at [Clinicaltrials.gov](https://clinicaltrials.gov) (NCT05685339) before the first recruitment, with [Figure 1](#) in [Supplementary Figure 1](#) illustrating trial design.

Between January 2023 and March 2024, participants were recruited and randomized across 3 research centers embedded in dialysis clinics located within a 50-km radius of the coordinating center. Eligible patients

were aged ≥ 18 years and were on regular dialysis regimens for > 3 months. Exclusion criteria included liver failure, pregnancy, allergy to study drug, and recent cardiovascular events (defined as decompensated HF or acute coronary syndrome within 3 months prior to screening). All participants were required to demonstrate clear capacity for informed consent and maintain a stable dialysis regimen, with no planned modifications within 6 months of enrollment.

Trial Design and Treatment

Participants were randomized 1:1 to receive dapagliflozin 10 mg orally once daily alongside standard treatment (dapagliflozin group) or standard treatment alone (control group) for 24 weeks, using a blocked simple randomization sequence generated electronically and managed through the Research Electronic Data Capture system. Film-coated dapagliflozin tablets were provided to the dapagliflozin group in sufficient number for uninterrupted treatment throughout study period. The control group maintained their standard treatment—defined as the prescription determined by their attending nephrologist—without interference from study personnel.

Assessments

Each research site oversaw screening, randomization, study drug distribution, and monthly assessment of adherence, and safety end points. In addition, they were responsible for keeping updated records on demographic data, medical history, and key dialysis parameters. Because hemodialysis sessions were scheduled at the participating centers on Mondays, Wednesdays, and Fridays, to minimize interexamination variability, standardized end point assessment occurred on midweek nondialysis days at the coordinating center (Thursday afternoon). At these visits, participants answered the KCCQ and received a comprehensive evaluation, including 12-lead electrocardiogram, blood sampling, vital signs, and anthropometric measurements. They also completed a standardized 6MWT and speckle tracking echocardiography.

The KCCQ patient-reported health status tool was answered by all participants.^{13,14} Using established scoring algorithms, each questionnaire item was rescaled and integrated into the following 4 distinct domains: physical limitation, symptom frequency, quality of life, and social limitation. The KCCQ-total summary scale reflected the mean score across all domains, whereas the KCCQ-clinical summary scale was calculated using the same methodology but excluded the quality-of-life domain. Aligned with clinical benchmarks, we defined the minimum clinically important difference as a change > 5 points in KCCQ-

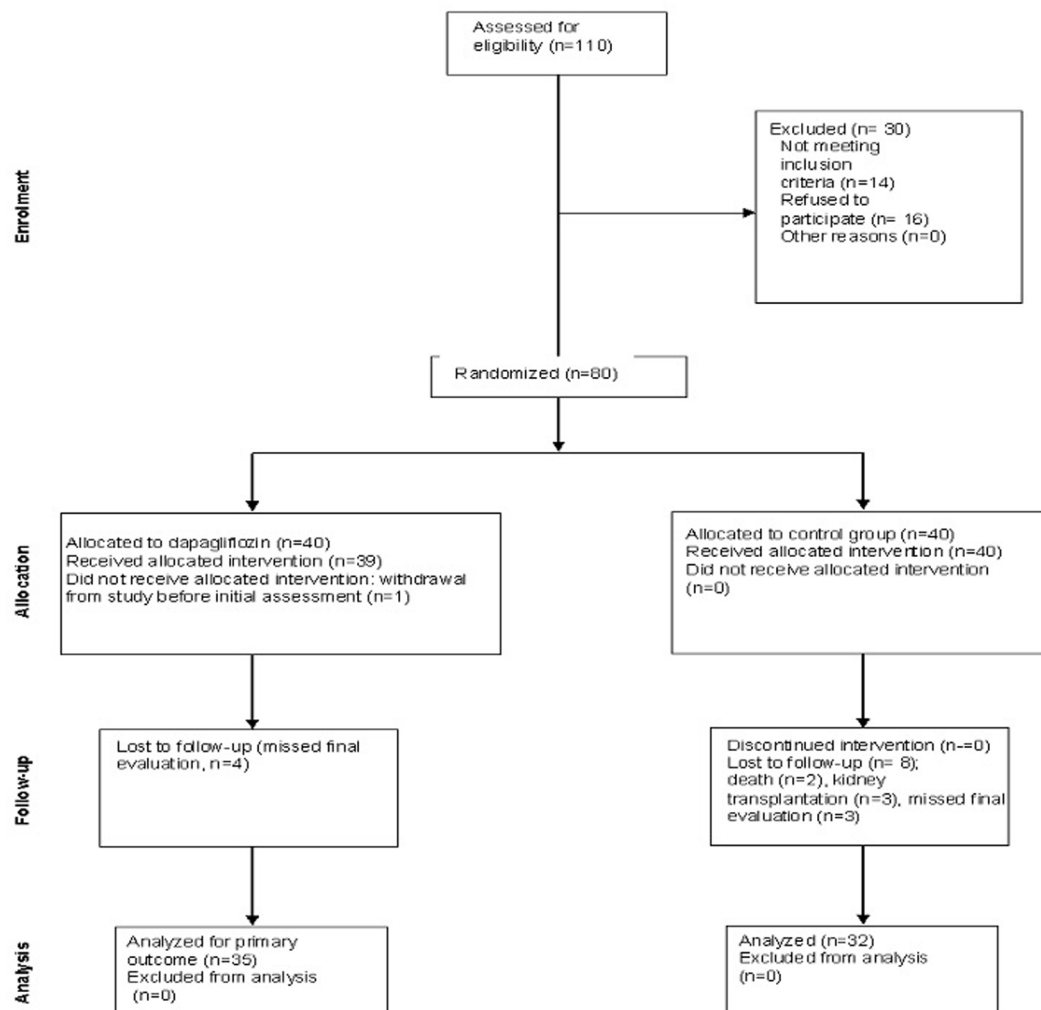


Figure 1. CONSORT flowchart.

12 scales, a validated threshold representing the smallest outcome change that patients perceive as meaningful and beneficial.¹⁵

The 6MWT complied with the American Thoracic Society guidelines.¹⁶ Participants walked at their maximum sustainable pace along a 30-m flat, enclosed corridor with a hard surface for exactly 6 minutes. Comprehensive vital signs were recorded at rest and immediately postexertion, using a calibrated automated device. Walking distance was documented using standardized corridor demarcations. Contraindications comprised impaired vision, limited mobility, resting systolic blood pressure > 180 mm Hg, and resting heart rate > 120 beats/min, safeguarding participant well-being and data integrity. Walking distance improvements > 15 m were deemed clinically relevant, because this threshold has demonstrated prognostic value in HF trials.¹⁷

Echocardiography complied with the American Society of Echocardiography guidelines and was performed using an ultrasound (EPIC CVX, Phillips

Medical Systems, ANDOVER, MA) equipped with 2.4 to 10 MHz linear array transducer (X5-1 xMATRIX; Philips Medical Systems, Andover, MA) by a trained cardiologist (SKM) blinded to treatment allocation.¹⁸ Evaluation of systolic function included left ventricle ejection fraction and left ventricle global longitudinal strain, whereas diastolic function appraisal considered E/e' ratio. Cardiac structure analysis incorporated indexed left ventricle mass and left atrial volume. HF diagnosis complied with the 2019 ESC guidelines recommendations.¹⁹

Upon study completion, investigators blinded to group allocation completed simultaneous batch analysis of cryopreserved plasma aliquots to ensure appropriate measurements of the following key biomarkers: NT-proBNP (Human NT Pro-BNP enzyme-linked immunosorbent assay, R&D Systems, NE, Minneapolis, MN), β -hydroxybutyrate (Colorimetric Assay Kit, Cayman Chemical, Ann Arbor, MI), and ultrasensitive cardiac troponin I (cTnI; Human High Sensitivity Cardiac Troponin I enzyme-linked

immunosorbent assay kit, MyBioSource, San Diego, CA). Blood sampling protocol is provided in Study Protocol ([Supplementary Material](#)).

Dialysis Regimens

Study patients were either on a standard hemodialysis regimen consisting of 3 sessions/wk, each lasting 4 hours, using the Dialog+ (B Braun, Germany) machine and high efficiency dialyzers (HF80S Dialyzer High Flux, Fresenius, Germany), or on automated peritoneal dialysis regimens of daily overnight sessions, lasting 8 hours each, using the HOMECHOICE Automated PD System (Baxter International Inc, USA). The adequacy of hemodialysis sessions was secured by regular single-pool Kt/V measurements; and prescription of ultrafiltration goal, dialysate concentration, and blood flow rate were individualized at each session by the attending nephrologist. The ultrafiltration volume of the last dialysis session prior to randomization was registered as a proxy of interdialytic weight gain. Dialysate prescription, during this study, was determined solely by the attending nephrologist's clinical judgment, with no interference from the study personnel.

End Points

The primary efficacy end point was the change from baseline in NT-proBNP levels at 24 weeks. Key secondary efficacy end points were 24-week changes in KCCQ scores and 6MWT. Other secondary end points included the number of patients with clinically meaningful changes in 6MWT or KCCQ scales. Exploratory end points comprised changes in speckle tracking echocardiography and cTnI measurements.

Safety outcomes encompassed both prespecified clinical events and laboratory parameters. Clinical events included cardiovascular death, HF hospitalization, myocardial infarction, symptomatic dehydration, euglycemic ketoacidosis, amputation, and drug-induced liver injury. Study sites were responsible for the diligent surveillance of safety end points during monthly follow-up visits or via participant reports. All confirmed or suspected adverse events were reported to the coordinating center to ensure the timely initiation of proportionate investigative measures. Laboratory safety parameters included hematologic indices (hemoglobin, hematocrit), electrolytes, liver enzymes, and β -hydroxybutyrate levels. Medication adherence was quantified as the percentage difference between observed and expected pills utilization.

Statistical Analysis

A statistical analysis plan was developed *a priori* and is presented in detail in [Supplementary Material](#).¹²

Briefly, for data presentation, we used standard conventions as follows: normally distributed continuous variables as mean (SD), nonnormally distributed continuous variables as median (interquartile range), and categorical variables as absolute numbers and percentages. For intergroup comparisons, we employed appropriate statistical methods as follows: *t* test for normally distributed continuous variables, Mann-Whitney U test for nonnormally distributed continuous variables, and chi-square test for categorical variables. All efficacy analysis adhered to the intention-to-treat principle.

For the primary analytical approach for continuous end points, we used analysis of covariance to determine intergroup LSD in mean change from baseline, with appropriate adjustment for baseline values. The difference between study arms in the median change, adjusted by baseline values using the Hodges Lehman method, in nonparametric end points was analyzed using the rank-based (Quade) analysis of covariance. For nonparametric continuous end points, the comparison of log-transformed changes from baseline was performed as sensitivity analysis when appropriate. No adjustment for multiple comparisons was applied to secondary or exploratory end points, which were analyzed descriptively. Interaction analysis was performed to assess the effect of baseline characteristics (age, sex, dialysis modality, dialysis length, residual diuresis, NT-proBNP levels, and HF status) on the primary outcome. Categorical secondary and/or safety end points were analyzed using logistic regression models to generate ORs with corresponding CIs (95% CI). All statistical analyses were performed using R software (R Foundation for Statistical Computing, Vienna, Austria).

Sample Size

We assumed that dapagliflozin would reduce NT-proBNP levels by 8% compared with standard treatment. For this effect size, a sample size of 52 patients would be required for 0.05 significance level and 80% statistical power. We aimed to enroll 40 participants per arm, anticipating attrition in this population. For sample size calculation, we used the G Power calculator (Heinrich Heine Universität Düsseldorf, Germany).

RESULTS

Enrollment process and participants flow are depicted in the CONSORT diagram ([Figure 1](#)). After the exclusion of 1 participant randomized to the dapagliflozin group who did not complete baseline assessments, the study population comprised 79 individuals assigned to dapagliflozin ($n = 39$) or control group ($n = 40$). The mean age was 56 ± 16 years, and 43 (54%) were male.

Hemodialysis was the modality for 67 patients (85%), 41 (52%) were anuric, and the median (interquartile range) dialysis duration was 3.5 (1.7–6.5) years. HF was diagnosed in 54 patients (68%) and hypertensive nephropathy was the cause of end-stage kidney disease in 30 patients (38%). Study medication adherence rate was $93\% \pm 13\%$ and follow-up time was 171 ± 31 days. No dialysis session was lost during the study period and there was no dialysis modality change over the study period. Clinical characteristics were well-balanced between study groups (Table 1). Sixty-seven patients (85%) completed the study treatment and were included in the primary analysis.

Primary End Point

In a rank-based analysis of covariance (Quade test) adjusting for baseline NT-proBNP, the between-group difference in NT-proBNP change was not statistically significant ($P = 0.065$). Although the unadjusted Hodges-Lehmann estimate suggested a median reduction of -155 pg/ml (95% CI: -327 to -33) with dapagliflozin compared with control, this effect was attenuated after accounting for baseline NT-proBNP levels (Table 2). Sensitivity analysis showed no interaction between predefined categories and the primary outcome (Figure 2). In sensitivity analysis, the intergroup difference in NT-proBNP was not statistically significant after log-transformation of the NT-proBNP levels change in each group ($P = 0.068$).

Secondary End Points

Dapagliflozin compared with the control group did not change symptom burden. No significant intergroup difference was reported for changes in KCCQ-total summary scale (LSD, 7 points [95% CI: -1 to 14]; $P = 0.094$) or KCCQ-clinical summary scale (LSD, 7 points [95% CI: -1 to 14]; $P = 0.073$) (Table 2). A total of 6 participants (18%) from the control group and 8 (22.9%) from the dapagliflozin group experienced a clinically meaningful improvement in KCCQ-clinical summary scale (OR: 1.63 [95% CI: 0.33–9.12]; $P = 0.548$). As shown in Table 2, no statistically significant intergroup difference was found with respect to KCCQ subdomains scores. The number of patients with clinically meaningful improvement in KCCQ-total summary scale was comparable between control ($n = 7$) and dapagliflozin ($n = 12$) groups (OR: 2.91 [95% CI: 0.74–14.06]; $P = 0.145$) (Table 3). KCCQ subdomain scores varied comparably between study arms (Tables 2 and 3).

The 24-week difference in 6MWT change was nonsignificant (LSD, -1 meter [95% CI: -34 to 32]; $P = 0.956$) (Table 2). 6MWT improved in 10 patients (34.5%) from the control group and in 12 patients

(40%) from the dapagliflozin group (OR: 1.27; 95% CI: 0.44–3.74; $P = 0.241$) (Table 3). There were no significant intergroup differences in speckle tracking echocardiography measures or cTnI over the study period (Table 2).

Safety Analysis

Overall, dapagliflozin exhibited a favorable safety profile in this dialysis population. Two deaths occurred in the control group (1 from abdominal sepsis and 1 from pulmonary sepsis), whereas no deaths were reported in the dapagliflozin group. Hospitalization events were infrequent and balanced: 2 in the dapagliflozin (1 for decompensated HF, 1 for transient ischemic stroke; 1 for osteomyelitis); none in the control group (Table 4). Symptomatic dehydration, euglycemic ketoacidosis, limb amputation, and drug-induced liver injury did not occur. Dapagliflozin therapy increased hematocrit levels, but not hemoglobin, electrolytes, liver enzymes, or β -hydroxybutyrate levels (Table 3).

DISCUSSION

This study is the first to address the hypothesis of cardiovascular benefit of SGLT2i in individuals under maintenance dialysis regimens. Our study did not demonstrate cardiovascular benefit in this setting, as evidenced by the absence of improvement in NT-proBNP levels, exercise capacity, symptom burden, or myocardial structure and function after 6-month therapy with dapagliflozin. This effect was irrespective of the dialysis modality.

As noted above, substantial evidence supports the use of SGLT2i to prevent the onset and progression of HF, including in individuals with impaired kidney function.⁶ Mechanisms unrelated to kidney effects, particularly those involving myocardial bioenergetics and vascular function, are thought to underlie the cardiovascular benefits of SGLT2i, independently of kidney function. Given the sharp increase in HF incidence following the initiation of dialysis and the absence of effective interventions to mitigate this risk, SGLT2 inhibition has emerged as a compelling therapeutic hypothesis warranting investigation in clinical trials. We previously demonstrated that dialysis does not alter the pharmacokinetics of dapagliflozin and is not associated with safety concerns.²⁰ The present trial was designed as a direct extension of that work, aiming to assess surrogate end points related to HF pathophysiology and clinical manifestation in this high-risk population.

The primary end point of this trial was the change in NT-proBNP levels. NT-proBNP levels are positively associated with the risk of cardiovascular death in

Table 1. Baseline characteristics of study population

Variable	Overall	Control	Dapagliflozin	P-value
Demographical				
Age, yrs	55.67 ± 15.69	57.55 ± 14.41	53.74 ± 16.87	0.285
Male	43 (54.4%)	21 (52.5%)	22 (56.4%)	0.902
Ethnicity				0.245
Black	18 (22.8%)	6 (15.0%)	12 (30.8%)	
Brown	13 (16.5%)	7 (17.5%)	6 (15.4%)	
Caucasian	48 (60.8%)	27 (67.5%)	21 (53.8%)	
Medical history				
Hemodialysis	67 (84.8%)	31 (77.5%)	36 (92.3%)	0.129
Dialysis length, yrs	3.52 (1.67–6.53)	3.48 (1.39–8.29)	3.52 (2.27–6.06)	0.624
GFR, ml/min per 1.73 m ²	6.7 (5.2–8.7)	6.5 (5.1–8.6)	6.8 (5.2–9.1)	0.133
Urinary output, ml	600 (400–1000)	1000 (500–1000)	500 (300–900)	0.785
Ultrafiltration volume, l	2.1 (1.5–2.6)	2.1 (1.6–2.6)	2.0 (1.5–2.6)	0.540
Interdialytic weight gain, kg	2.0 (1.4–2.5)	2.0 (1.5–2.5)	1.9 (1.4–2.5)	0.613
Anuria	38 (48.1%)	19 (47.5%)	19 (48.7%)	1.000
Smoker	6 (7.6%)	1 (2.5%)	5 (12.8%)	0.191
Prior myocardial infarction	12 (15.2%)	3 (7.5%)	9 (23.1%)	0.106
Prior stroke	1 (1.3%)	0 (0.0%)	1 (2.6%)	0.990
Atrial fibrillation	1 (1.3%)	0 (0.0%)	1 (2.6%)	0.990
Diabetes	14 (17.7%)	5 (12.5%)	9 (23.1%)	0.349
Hypertension	58 (73.4%)	29 (72.5%)	29 (74.4%)	1.000
Heart failure	54 (68%)	25 (63%)	29 (75%)	0.260
Prior kidney transplantation	8 (10.1%)	4 (10.0%)	4 (10.3%)	1.000
Kidney disease etiology				
Hypertensive	30 (38.0%)	14 (35.0%)	16 (41.0%)	0.650
Diabetic	12 (15.2%)	4 (10.0%)	8 (20.5%)	0.210
Glomerulonephritis	9 (11.4%)	5 (12.5%)	4 (10.3%)	1.000
Polycystic kidney disease	8 (10.1%)	4 (10.0%)	4 (10.3%)	1.000
Obstructive nephropathy	6 (7.6%)	5 (12.5%)	1 (2.6%)	0.360
NSAID	5 (6.3%)	3 (7.5%)	2 (5.1%)	1.000
AKI	4 (5.1%)	1 (2.5%)	3 (7.7%)	0.620
Unknown	5 (6.3%)	4 (10.0%)	1 (2.6%)	0.360
Physical examination				
Heart Rate, bpm	77 ± 13	78 ± 13	77 ± 14	0.761
Systolic BP, mm Hg	140 ± 27	142 ± 30	138 ± 25	0.591
Diastolic BP, mm Hg	82 ± 18	85 ± 21	79 ± 13	0.134
Dry Weight, kg	72 ± 14	72 ± 14	71 ± 15	0.698
Body mass index, kg/m ²	26 ± 5	26 ± 5	25 ± 4	0.246
Laboratory tests				
Hemoglobin, g/dl	12.6 ± 1.9	12.8 ± 1.7	12.4 ± 2.2	0.332
Hematocrit, %	38 ± 6	38 ± 56	37 ± 7	0.509
Fasting blood glucose, mg/dl	86 (70–103)	81 (71–101)	88 (68–107)	0.831
Glycated hemoglobin, %	5.1 (4.8–5.4)	5.1 (4.8–5.5)	5.2 (4.8–5.4)	0.732
Urea, mg/dl	87 (73–108)	89 (74–120)	85 (69–102)	0.121
Creatinine, mg/dl	7.39 (6.07–9.79)	7.46 (6.00–10.29)	7.30 (6.07–9.69)	0.906
Sodium, mEq/l	137 ± 4	137 ± 4	137 ± 3	0.964
Potassium, mEq/l	5.1 ± 0.8	5.1 ± 0.8	5.0 ± 0.8	0.668
AST, mg/dl	13 (11–17)	13 (11–22)	13 (10–16)	0.281
ALT, mg/dl	12 (9–18)	13.00 (10–20)	12 (8–15)	0.077
pH	7.30 ± 0.08	7.30 ± 0.07	7.30 ± 0.08	0.928
Bicarbonate, mg/dl	25 ± 4	24 ± 4	27 ± 5	0.006
NT-proBNP, pg/ml	626 (196–1400)	316 (180–1426)	747 (247–1400)	0.251
Troponin, ng/ml	383 (248–876)	361 (246–961)	448 (248–815)	0.871
Beta-hydroxybutyrate, mmol/L	0.36 ± 0.09	0.34 ± 0.05	0.38 ± 0.11	0.098
Medications				
Thiazides	6 (7.6%)	1 (2.5%)	5 (12.8%)	0.191
Furosemide	34 (43.0%)	16 (40.0%)	18 (46.2%)	0.745
Betablocker	42 (53.2%)	17 (42.5%)	25 (64.1%)	0.089
ARB or ACEi	29 (36.7%)	15 (37.5%)	14 (35.9%)	1.000

(Continued on following page)

Table 1. (Continued) Baseline characteristics of study population

Variable	Overall	Control	Dapagliflozin	P-value
Calcium channel blocker	17 (21.5%)	4 (10.0%)	13 (33.3%)	0.024
Hydralazine	11 (13.9%)	3 (7.5%)	8 (20.5%)	0.179
Clonidine	3 (3.8%)	2 (5.0%)	1 (2.6%)	1.000
Espironolactone	2 (2.5%)	0 (0.0%)	2 (5.1%)	0.463
Insulin	7 (8.9%)	3 (7.5%)	4 (10.3%)	0.972
Calcium carbonate	16 (20.3%)	7 (17.5%)	9 (23.1%)	0.736
Colecalciferol	18 (22.8%)	12 (30.0%)	6 (15.4%)	0.200
Calcitriol	7 (8.9%)	6 (15.0%)	1 (2.6%)	0.121
Omeprazole	13 (16.5%)	5 (12.5%)	8 (20.5%)	0.511
Anticoagulant	2 (2.5%)	0 (0.0%)	2 (5.1%)	0.463
Corticoid	5 (6.3%)	4 (10.0%)	1 (2.6%)	0.371
Statin	15 (19.0%)	5 (12.5%)	10 (25.6%)	0.229
Iron	14 (17.7%)	6 (15.0%)	8 (20.5%)	0.729
Erythropoietin	23 (29.1%)	13 (32.5%)	10 (25.6%)	0.672
Sevelamer	47 (59.5%)	26 (65.0%)	21 (53.8%)	0.435
Cinacalcet	25 (31.6%)	13 (32.5%)	12 (30.8%)	1.000
6-minute walk test distance	403 ± 102	407 ± 93	399 ± 111	0.750
Echocardiography				
LVEDV, ml	117 ± 32	111 ± 28	123 ± 34	0.097
LVESV, ml	47 ± 26	41 ± 16	53 ± 32	0.043
LVEF, %	61 ± 14	64 ± 9	58 ± 17	0.068
LVMi, g/m ²	106 ± 39	102 ± 34	111 ± 43	0.301
E/e ⁻ ratio	8.7 (7.1–12.8)	8.9 (7.2–12.8)	8.7 (7.0–14.2)	0.885
LV-GLS, %	−19 ± 5	−20 ± 5	−19 ± 5	0.554
LAVi, ml/m ²	32 ± 12	30 ± 12	33 ± 13	0.312
KCCQ domains				
Total summary score	84 ± 23	84 ± 25	85 ± 21	0.861
Clinical summary score	86 ± 22	85 ± 24	87 ± 20	0.727
Physical limitation	84 ± 26	84 ± 29	83 ± 24	0.951
Symptom frequency	85 ± 23	82 ± 25	87 ± 21	0.373
Quality of life	79 ± 29	79 ± 31	80 ± 25	0.869
Social limitation	89 ± 23	90 ± 25	88 ± 20	0.766

ACEi, angiotensinogen-converting enzyme inhibitor; AKI, acute kidney injury; ALT, alanine transferase; ARB, angiotensin receptor blocker; AST, aspartate aminotransferase; BP, blood pressure; FAC, fractional area curve; GFR, glomerular filtration rate; KCCQ, Kansas City Cardiomyopathy Questionnaire; LAVi, body surface area-indexed left atrial volume; LVEDV, left ventricular end-diastolic volume; LVEF, left ventricular ejection fraction; LVESV, left ventricular end-systolic volume; LV-GLS, left ventricular global longitudinal strain; LVMi, body-surface area-indexed left ventricular mass; NSAID, nonsteroidal antiinflammatory drug; NT-proBNP, N-terminal pro-brain natriuretic peptide; PSAP, pulmonary systolic arterial pressure; TAPSE, tricuspid annular plane systolic excursion.

Data presented as mean ± SD, median (interquartile range) or count (frequency, %). P-value compared the values of control with dapagliflozin group. Abbreviations.

individuals undergoing maintenance hemodialysis, and reductions in NT-proBNP have been linearly correlated with lower incidence of sudden cardiac death.²¹ In HF trials, SGLT2i have consistently lowered NT-proBNP levels, with greater reductions associated with improved clinical outcomes.²² With respect to dapagliflozin, 8 months therapy was associated with decrease in NT-proBNP by 303 pg/mL.²³ Collectively, these findings support the hypothesis that NT-proBNP levels in dialysis patients would be favorably modified by dapagliflozin. However, this hypothesis was not confirmed by our trial.

The 6MWT parameters did not change in either group in our trial. Although dapagliflozin may confer modest improvements in exercise capacity in certain subgroups of patients with HF, its effect on 6MWT has not been consistently significant across studies. A meta-analysis suggested a potential benefit in HF with

preserved ejection fraction, with a mean increase of 10 m.²⁴ However, the DETERMINE trial, which included patients with both HF with reduced ejection fraction and those with HF with preserved ejection fraction, found no significant effect of dapagliflozin on 6MWT.²⁵ The lack of therapeutic benefit on this marker in our study is consistent with the neutral effect on the primary outcome, though generalizability of our findings still warrants acknowledgment of sample size limitation relative to previous studies.

Dapagliflozin has been shown to improve health status, as assessed by the KCCQ-12, particularly in patients with HF with reduced ejection fraction and, to a lesser extent, in those with HF with preserved ejection fraction.^{25,26} In our study, the change in KCCQ-12 scores over 24 weeks did not differ between treatment arms, nor did the incidence of clinically meaningful improvement. Subdomain scores changed similarly in

Table 2. Continuous end points analysis

Variable	Control (change)	Dapagliflozin (change)	Treatment effect	p-value
NT-proBNP, pg/ml	13 (–64 to 281)	–109 (–356 to 55)	–155 (–327 to –33)	0.065
KCCQ				
Total summary score	–3 ± 22	4 ± 16	7 (–1 to 14)	0.094
Clinical summary score	–2 ± 20	4 ± 15	7 (–1 to 14)	0.073
Physical limitation	–2 ± 24	3 ± 16	5 (–3 to 14)	0.194
Symptom frequency	1 ± 27	5 ± 18	7 (–1 to 15)	0.084
Quality of life	–5 ± 30	1 ± 22	7 (–4 to 18)	0.196
Social limitation	–5 ± 20	5 ± 20	8 (–1 to 16)	0.070
6MWT, m	0.4 ± 66.7	–0.5 ± 62.6	–0.9 (–34.1 to 32.3)	0.956
Echocardiography				
LVEDV, ml	5.6 ± 18.3	–5.8 ± 25.9	–10.3 (–21.7 to 1.2)	0.078
LVESV, ml	1.4 ± 10.4	–1.0 ± 16.4	–1.0 (–8.1 to 6.0)	0.771
LVEF, %	0.4 ± 8.8	–0.1 ± 14.1	–3.5 (–8.8 to 1.9)	0.197
LVMi, g/m ²	1.0 ± 21.3	–3.7 ± 23.4	–1.7 (–12.8 to 9.5)	0.765
E/e' ratio	–0.3 (–1.8 to 1.0)	0.2 (–1.2 to 1.8)	–0.7 (–2.0 to 0.5)	0.278
LV-GLS, %	–0.1 ± 5.4	–0.7 ± 5.0	–0.2 (–2.5 to 2.1)	0.880
LAVi, ml/m ²	0.3 ± 8.1	1.6 ± 8.8	2.2 (–1.9 to 6.4)	0.284
Laboratory				
Hemoglobin, g/dl	–0.7 ± 2.0	0.5 ± 1.8	0.8 (0.0 to 1.7)	0.044
Hematocrit, %	–1.9 ± 6.4	1.7 ± 6.2	2.8 (0.2 to 5.4)	0.036
Glycated hemoglobin, %	0.0 (–0.2 to 0.2)	0.1 (0.0 to 0.3)	–0.1 (–0.2 to 0.0)	0.152
AST, mg/dl	1.0 (–3.0 to 3.5)	1.0 (–1.0 to 4.0)	–1.0 (–4.0 to 1.0)	0.475
ALT, mg/dl	1.0 (–4.5 to 3.5)	1.0 (–2.0 to 5.0)	–1.0 (–4.0 to 2.0)	0.962
pH	0.0 ± 0.1	0.0 ± 0.1	–0.0 (–0.0 to 0.0)	0.908
Bicarbonate, mEq/l	1.2 ± 4.2	0.2 ± 3.4	0.3 (–1.5 to 2.0)	0.767
Troponin, ng/ml	–3.3 (–67.8 to 40.2)	6.8 (–82.2 to 162.8)	–11.4 (–108.7 to 61.6)	0.556
Beta-hydroxybutyrate, mmol/l	–0.0 ± 0.1	–0.0 ± 0.2	0.0 (–0.0 to 0.1)	0.054

6MWT, 6-minute walk test distance; ALT, alanine aminotransferase; AST, aspartate aminotransferase; CI, confidence interval; KCCQ, Kansas City Cardiomyopathy Questionnaire; LAVi, body-surface area-indexed left atrial volume; LVEDV, left ventricle end-diastolic volume; LVEF, left ventricular ejection fraction; LVESV, left ventricular end-systolic volume; LV-GLS, LV global longitudinal strain; LVMi, body surface area-indexed LV mass; NT-proBNP, N-terminal pro-brain natriuretic peptide.

Baseline data are median (interquartile range), whereas change is presented as baseline-adjusted Hodges Lehman median (95% CI) change. Treatment effect is the Hodges Lehman median (95% CI) difference between dapagliflozin and control groups, with *P*-value representing the results of the rank-based analysis of covariance.

the dapagliflozin and control groups. Although the KCCQ-12 has demonstrated sensitivity to early symptomatic changes in CKD, even in the absence of HF, its psychometric properties have not been validated in dialysis. Notably, the overlap of the clinical manifestations of end-stage kidney disease with HF might have attenuated the capacity of KCCQ-12 to disentangle the effect of dapagliflozin on physical and psychosocial stressors, arising from dialysis treatment, that may have been overlooked by questionnaire items.²⁷ This drawback may have limited our study's capacity to provide definitive answer regarding symptomatic improvement.

Our findings suggest that a minimal level of residual kidney function may be necessary for dapagliflozin to exert its beneficial effects on HF-related end points. In individuals with preserved kidney function, dapagliflozin rapidly promotes urinary sodium and glucose excretion, resulting in a mean increase of approximately 300 ml in daily urine output and a reduction in both interstitial and total body water.²⁸ These changes lead to acute reductions in myocardial stretch and symptoms of fluid overload.²⁶ Notably, in individuals without

end-stage kidney disease, significant improvements in NT-proBNP levels and KCCQ scores have been observed as early as 1 month after initiation of therapy.^{23,26,29}

Increased urinary glucose excretion promotes a metabolic shift in the myocardium toward ketone body utilization, thereby enhancing myocardial contractility.^{30,31} In a crossover trial, dapagliflozin led to a 400% relative increase in circulating β -hydroxybutyrate compared with placebo.³⁰ Based on such findings, it has been proposed that enhanced ketogenesis may contribute to the cardioprotective effects of SGLT2i. If so, the loss of urinary glucose excretion and dialytic clearance of ketone bodies in patients undergoing dialysis may attenuate this mechanism of benefit. Supporting this notion, we observed no change in β -hydroxybutyrate levels in dialytic patients treated with dapagliflozin in the present study.

Dapagliflozin resulted in a 2.7% relative increase in hematocrit levels in our trial. This finding is consistent with increases observed in the DAPA-CKD (2.3%)³² and DECLARE-TIMI 58 (2.5%)³³ trials. Although initially attributed to hemoconcentration secondary to volume contraction, emerging preclinical evidence

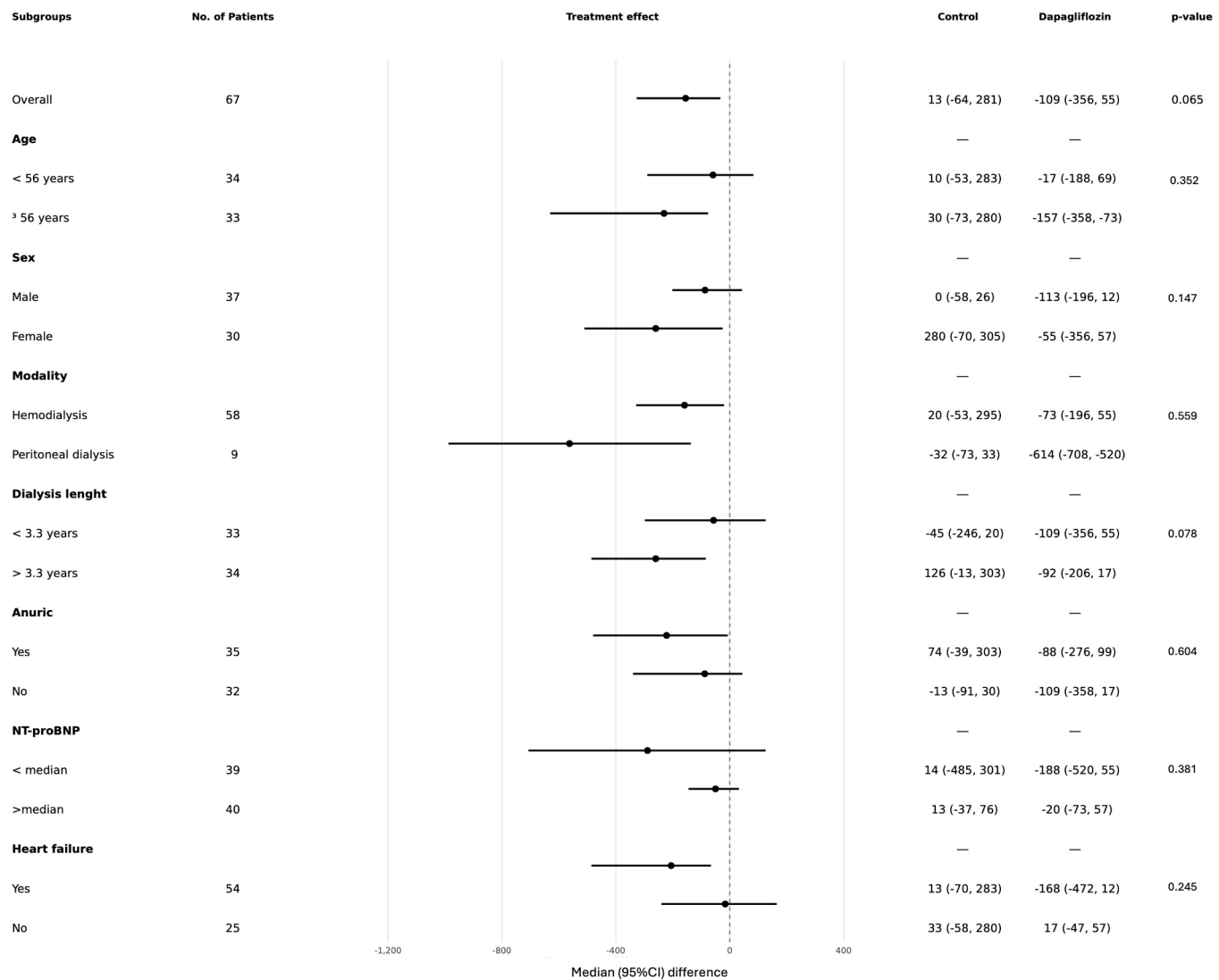


Figure 2. Primary efficacy end point subgroup analysis. NT-proBNP, N-terminal pro-brain natriuretic peptide.

Table 3. Categorical end points analysis

Outcome	n/total Control	n/total Dapa	Adjusted OR (95% CI)	P-value
Clinically meaningful improvement				
KCCQ-CSS	6/32 (18.8%)	8/35 (22.9%)	1.63 (0.33 to 9.12)	0.548
KCCQ-TSS	7/32 (21.9%)	12/35 (34.3%)	2.91 (0.74 to 14.06)	0.145
6-MWTD	10/29 (34.5%)	12/30 (40.0%)	1.27 (0.44 to 3.74)	0.657
Any improvement				
KCCQ-CSS	9/32 (28.1%)	13/35 (37.1%)	1.79 (0.54 to 6.41)	0.348
KCCQ-TSS	9/32 (28.1%)	14/35 (40.0%)	2.48 (0.68 to 10.51)	0.185
Physical limitation	6/32 (18.8%)	13/35 (37.1%)	7.87 (1.35 to 86.08)	0.042
Symptom frequency	10/32 (31.2%)	9/35 (25.7%)	1.02 (0.23 to 4.75)	0.981
Quality of life	8/32 (25.0%)	12/35 (34.3%)	2.31 (0.60 to 10.39)	0.241
Social limitation	3/32 (9.4%)	7/35 (20.0%)	2.82 (0.40 to 28.39)	0.319
6MWT distance	12/29 (41.4%)	19/30 (63.3%)	2.51 (0.88 to 7.48)	0.090

6MWT, 6-minute walk test distance; CI, confidence interval; CSS, clinical summary scale; KCCQ, Kansas City Cardiomyopathy Questionnaire; OR, odds ratio; TSS, total summary scale.

suggests a predominant mechanism involving stimulation of erythropoiesis through upregulation of erythropoietin and erythropoietin, as well as inhibition of hypoxia-inducible factor-1 α .³⁴ These pathways enhance both kidney and hepatic erythropoiesis, the latter of which remains preserved in patients with end-stage kidney disease and may thus account for the observed increase in hematocrit in this population. Whether hematocrit increase from dapagliflozin yields long-term clinical benefits, for example through improved oxygen-carrying capacity and less exertional and interdialytic fatigue, warrants further investigation.

The interpretation of our findings warrants acknowledgment of potential limitations. As the first study evaluating the effect of dapagliflozin on NT-proBNP in chronic dialysis, the anticipated treatment

Table 4. Safety outcomes

Categorical safety outcomes	Control	Dapagliflozin
Any safety event	2	3
Amputation	0	0
Major cardiovascular event	0	1
Cardiovascular death	0	0
Death (noncardiovascular cause)	2	0
Diabetic ketoacidosis	0	0
Genital infection	0	0
Hospitalization (any cause)	0	0
Hospitalization for heart failure	0	1
Hypoglycemia	0	0
Myocardial infarction	0	0
Osteomyelitis	0	1
Stroke (ischemic or hemorrhagic)	0	0
Transient ischemic attack	0	1
Urinary tract infection	0	0
Volume depletion/hypotension	0	0

effect for sample size calculation was based on previous reports derived from clinical trials that excluded this population. If dialysis, through fluid fluctuations and dialysate loss, influence the capacity of dapagliflozin to modify this end point, then the forecasted treatment effect was overestimated, thereby limiting statistical power. This shortcoming may have limited the precision to confirm clinically meaningful benefits of dapagliflozin on cardiac remodeling. The demonstration of NT-proBNP reductions from dapagliflozin treatment across study subgroups, though not statistically significant, should hence be interpreted judiciously.

This study was susceptible to limitations introduced by the open label design. Though observer bias was attenuated by end point assessment blinding, this approach is insufficient to prevent expectation bias in subjective end points. As a result, it is possible that through expectation bias, treatment effect on symptom perception was exaggerated, which might have contributed to numerically higher scores in KCCQ-12 domains, despite no objective evidence of cardiovascular benefit. Supporting this preposition, the difference between study arms in KCCQ domains were, on average, higher than that reported by the DAPA-CKD study, which may in part reflect differences in study design, including double-blinding.³ The open-label design may also have contributed to a higher proportion of participants in the control group missing the final evaluation, potentially reflecting perceived lack of benefit from study participation in addition to logistical burdens related to time commitment and travel to the research center. Completion rate imbalance in this context may have represented an additional source of systemic bias limiting generalizability of our results.

The uneven distribution of some key characteristics among participants represents a significant limitation. In this matter, the study sample was not enriched for HF with reduced ejection fraction relative to preserved ejection fraction, thereby limiting the extrapolation of our results to the phenotype that could potentially benefit the most from the intervention. Further, the disproportionately low number of patients receiving peritoneal dialysis relative to those receiving hemodialysis, as well as the lack of dialysis prescription standardization, precludes definitive assessment of effect modification by these features. This shortcoming is only partially addressed by subgroup analysis, which warrants recognition of potentially reduced statistical power. Finally, the interpretation of secondary and exploratory end points should account for the lack of multiplicity adjustment, reinforcing the reduced power to provide definitive answers concerning those metrics.

In summary, our study for the first time demonstrated that initiating dapagliflozin in patients on chronic dialysis is safe but did not influence NT-proBNP levels, functional capacity, or myocardial function or structure. The previously described stimulation of erythropoiesis was confirmed, as was the favorable safety profile of dapagliflozin in this population. Although interpretation of our findings warrants acknowledgement of sample size and open label design limitations, collectively this study shows that, although well-tolerated, dapagliflozin may not confer cardiovascular benefits when initiated in patients receiving dialysis.

CONCLUSION

Dapagliflozin was safe but did not affect surrogate markers of HF compared with standard care in patients receiving maintenance dialysis, suggesting no demonstrable cardiovascular benefit over 24 weeks in this population.

DISCLOSURE

All the authors declared no competing interests.

Funding

The study was designed and proposed by the BHS group and funded by FAPESP (Grant # 2023/16961-1). The BHS group assumed full legal and regulatory responsibility for planning, conducting, and managing the study in compliance with applicable Brazilian regulations and laws. Prof. Sposito was supported by a Research Career Awards grant (No. 304257/2021-4) from the Brazilian National Research Council (CNPq).

The findings and conclusions in this report are those of the authors and do not necessarily represent the official position of the funding institutions.

AUTHOR CONTRIBUTIONS

Concept and design were by STK-M and ACS. Acquisition, analysis, or interpretation of data were by all authors. Drafting of the manuscript was by STK-M and ACS. Critical review of the manuscript for important intellectual content was by all the authors. Statistical analysis was done by STK-M and ACS. Funding was obtained by ACS. Administrative, technical, or material support was provided by ACS. Supervision was by ACS.

DATA AVAILABILITY STATEMENT

The data supporting the findings of this study is openly available at the Unicamp Institutional Research Data Repository (<https://doi.org/10.25824/redu/OXTXSM>).

SUPPLEMENTARY MATERIAL

[Supplementary File \(PDF\)](#)

Supplementary Methods: Study Protocol.

Figure S1. Trial design.

CONSORT checklist.

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